

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 03: TISSUES: PLANT & ANIMAL STRUCTURES

TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 Which meristematic tissue is responsible for regenerating clipped grass blades after lawn-mowing or grazing?

- (A) Apical meristem (B) Intercalary meristem
(C) Cork cambium (D) Lateral meristem

Q.2 What complex aromatic polymer impregnates the secondary cell walls of sclerenchyma, imparting compressive strength and waterproofing?

- (A) Pectin (B) Lignin
(C) Cellulose (D) Suberin

Q.3 Which type of xylem vessel perforation facilitates rapid, bulk longitudinal water flow in angiosperms compared to gymnosperms?

- (A) Pitted imperforate tracheids (B) Perforation plates at vessel element end walls
(C) Suberin casparian strips (D) Sieve pore fields

Q.4 What is the primary histological function of companion cells in angiosperm phloem?

- (A) Mechanical support via thick lignin walls (B) Active loading and unloading of sugars into enucleated sieve tube elements via proton symport
(C) Water storage (D) Secretion of latex

Q.5 Which epidermal modification in roots drastically increases total absorptive surface area for water and mineral uptake?

- (A) Trichomes (B) Root hairs (unicellular extensions)
(C) Stomata (D) Cuticle

Q.6 Which tissue lines the human urinary bladder, allowing dramatic stretching and distension as urine accumulates?

- (A) Transitional Epithelium (Urothelium) (B) Ciliated Columnar
(C) Simple Squamous (D) Stratified Keratinized

Q.7 The microscopic structural functional unit of compact mammalian bone tissue is the:

- | | |
|-------------------------------|------------------------|
| (A) Osteon (Haversian System) | (B) Chondrocyte lacuna |
| (C) Sarcomere | (D) Nephron |
-

Q.8 Which non-collagenous protein fiber imparts reversible elastic stretchability to ligaments and arterial walls?

- | | |
|-------------|-------------|
| (A) Keratin | (B) Elastin |
| (C) Fibrin | (D) Myosin |
-

Q.9 Which cell type in human blood develops into antibody-secreting plasma cells during adaptive immune responses?

- | | |
|-------------------|-----------------|
| (A) B-Lymphocytes | (B) Neutrophils |
| (C) Erythrocytes | (D) Platelets |
-

Q.10 What is the primary organic matrix protein synthesized by chondrocytes in hyaline cartilage?

- | | |
|--|--------------------|
| (A) Type II Collagen and Chondroitin sulfate proteoglycans | (B) Hydroxyapatite |
| (C) Keratin | (D) Myosin |
-

Q.11 Which muscular tissue displays syncytial multinucleate fibers with peripherally located nuclei?

- | | |
|---------------------|-------------------|
| (A) Skeletal muscle | (B) Smooth muscle |
| (C) Cardiac muscle | (D) Myoepithelium |
-

Q.12 The insulating lipid-rich sheath surrounding many vertebrate nerve axons that accelerates impulse conduction is the:

- | | |
|-------------------|-----------------|
| (A) Myelin sheath | (B) Sarcolemma |
| (C) Perichondrium | (D) Perineurium |
-

Q.13 The gaps along a myelinated axon where voltage-gated sodium channels concentrate are called:

- | | |
|----------------------|------------------------|
| (A) Nodes of Ranvier | (B) Synapses |
| (C) Dendritic spines | (D) Intercalated discs |
-

Q.14 Which plant ground tissue contains cells that are dead at maturity and possess branched stone-cell morphologies (sclereids)?

- | | |
|-----------------|------------------------------------|
| (A) Collenchyma | (B) Sclerenchyma (Brachysclereids) |
| (C) Aerenchyma | (D) Chlorenchyma |
-

Q.15 The Casparian strip in the plant root endodermis is impregnated with which hydrophobic substance to force water into symplastic entry?

- | | |
|---------------|------------|
| (A) Suberin | (B) Cutin |
| (C) Cellulose | (D) Pectin |
-

Q.16 Which white blood cell type is the most abundant phagocyte in circulating human blood, arriving first at bacterial infection sites?

- | | |
|-----------------|-----------------|
| (A) Neutrophils | (B) Eosinophils |
| (C) Basophils | (D) Monocytes |
-

Q.17 Mast cells located in areolar connective tissue secrete which vasoactive substance responsible for inflammatory vasodilation and allergies?

- | | |
|---------------|----------------|
| (A) Histamine | (B) Insulin |
| (C) Pepsin | (D) Hemoglobin |
-

Q.18 Which specialized epithelial tissue possesses goblet cells that secrete protective viscous mucus?

- (A) Columnar epithelium (e.g., intestinal lining) (B) Squamous epithelium
(C) Dense regular tissue (D) Stratified cuboidal

Q.19 The contractile unit of a muscle myofibril bounded by two adjacent Z-lines is the:

- (A) Sarcomere (B) Sarcolemma
(C) Osteon (D) Synapse

Q.20 Which blood plasma protein is essential for maintaining intravascular colloidal osmotic (oncotic) pressure?

- (A) Serum Albumin (B) Fibrinogen
(C) Gamma globulin (D) Hemoglobin

Q.21 In woody stems, lenticels are specialized aerating pores located in the:

- (A) Bark (Periderm) (B) Woody xylem core
(C) Leaf petiole (D) Root tip

Q.22 Which cellular component of blood releases serotonin and thromboxane \$A_2\$ to cause localized vasoconstriction following vascular laceration?

- (A) Platelets (B) RBCs
(C) T-cells (D) Plasma albumin

Q.23 Chondrocytes are located within fluid-filled microscopic matrix cavities called:

- (A) Lacunae (B) Haversian canals
(C) Canaliculi (D) Ventricles

Q.24 Which muscle type contracts automatically under the control of pacemaker nodes without requiring conscious neural input?

- (A) Cardiac muscle (Sinoatrial node pacing) (B) Skeletal muscle
(C) Tongue muscle (D) Diaphragm muscle

Q.25 Which epithelial type forms the dry, impermeable, abrasion-resistant outer epidermis of mammalian skin?

- (A) Stratified Keratinized Squamous Epithelium (B) Simple Cuboidal
(C) Ciliated Columnar (D) Pseudostratified

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(B)	Intercalary meristems located at the bases of internodes and leaf blades divide actively to regenerate lost foliage.
Q.2	(B)	Lignin is a rigid polyphenolic polymer that reinforces xylem and sclerenchyma cell walls.
Q.3	(B)	Angiosperm vessel elements have completely open or scalariform perforation plates, providing low-resistance conduits.
Q.4	(B)	Companion cells utilize H^+ -ATPase symporters to actively load sucrose into sieve tube elements.
Q.5	(B)	Root hairs are delicate tubular outgrowths of root epidermal (epiblema) cells that amplify surface area.
Q.6	(A)	Transitional epithelium has multilayered rounded cells that flatten upon distension without tearing.
Q.7	(A)	Compact bone is organized into concentric cylindrical osteons centered on neurovascular Haversian canals.
Q.8	(B)	Elastin forms branched elastic fibers capable of stretching up to 1.5 times their length and snapping back.
Q.9	(A)	B-lymphocytes recognize specific antigens and differentiate into plasma cells that secrete circulating antibodies.
Q.10	(A)	Cartilage matrix consists of Type II collagen fibrils embedded in hydrated chondroitin sulfate proteoglycan gel.
Q.11	(A)	Skeletal muscle fibers arise from fusion of embryonic myoblasts, creating long syncytial multinucleated fibers.
Q.12	(A)	Schwann cells (PNS) and oligodendrocytes (CNS) produce myelin sheaths enabling saltatory conduction.
Q.13	(A)	Nodes of Ranvier are periodic gaps in myelin where action potentials regenerate rapidly via saltatory conduction.
Q.14	(B)	Sclereids (grit cells in pear fruit) are short, thick, lignified sclerenchyma cells providing compressive hardness.
Q.15	(A)	Suberin deposition in endodermal radial walls forms the Casparian strip, blocking apoplastic water movement.
Q.16	(A)	Neutrophils make up 60–70% of circulating WBCs and act as primary phagocytes during acute bacterial inflammation.
Q.17	(A)	Mast cells degranulate to release histamine and heparin during allergic reactions and tissue injury.
Q.18	(A)	Simple columnar epithelium of the stomach and intestine contains goblet cells that secrete protective mucus.
Q.19	(A)	The sarcomere is the basic repeating structural and functional contractile unit containing actin and myosin filaments.
Q.20	(A)	Albumin constitutes ~60% of total plasma protein and prevents fluid from leaking out of capillaries into tissues.
Q.21	(A)	Lenticels are loose, non-suberized cellular openings in tree bark that permit gas exchange in mature woody organs.
Q.22	(A)	Activated platelets release vasoconstrictors to minimize blood loss while building the primary hemostatic plug.
Q.23	(A)	Living cartilage cells (chondrocytes) occupy isolated spaces within the extracellular matrix termed lacunae.
Q.24	(A)	Cardiac muscle possesses myogenic autorhythmicity initiated by the heart's natural pacemaker (SA node).
Q.25	(A)	Keratinized stratified squamous epithelium consists of dead superficial cellular layers packed with tough insoluble keratin.